WORKING SCIENTIFICALLY

Science Investigators

Testable questions, predictions, fair tests, measuring, repeating, tables and bar charts


magnifying glass


test tube


stopwatch


bar chart

In this pack

Learn it: Science investigators ★☆☆
Fact check: Tools, units and tests ★☆☆
Think it through: Making it fair ★★☆
Investigate: Which paper makes the best paper plane? ★★☆
Real-world case study: Testing umbrellas for the monsoon ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · W1 · WORKING SCIENTIFICALLY

Science investigators

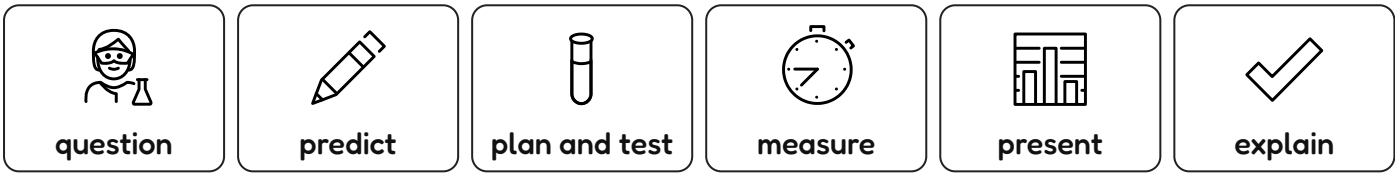
★☆☆ Grade 3

Name: Class: Date:

Learning goals

- Ask questions that can be tested and make predictions.
- Plan a fair test, measure carefully and repeat measurements.
- Show results in tables and bar charts and say what they show.

Investigators find out the truth by collecting **evidence**. Scientists are investigators of the natural world. They ask questions, plan tests, measure carefully and use evidence to explain what they find.



Questions you can test

"Which paper makes the best paper plane?" can be **tested**. "Which colour is prettiest?" cannot, because it is just an opinion.

Predict

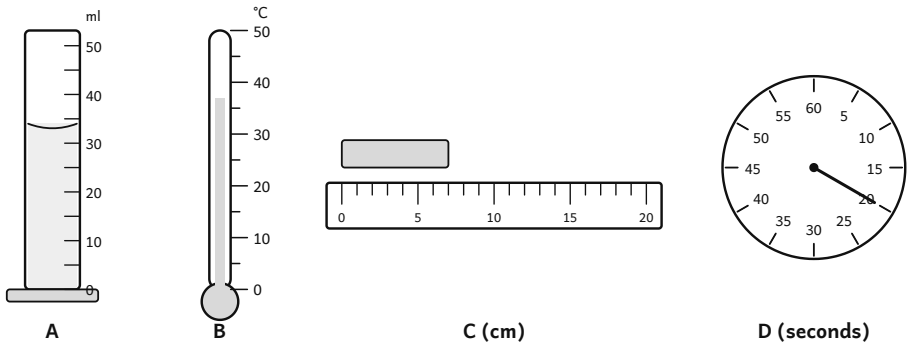
Use what you know: "I think the heavy paper plane will fly furthest **because** it will not be blown about so easily."

Plan a fair test

Change one thing	the type of paper
Measure one thing	how far the plane flies (in metres)
Keep the same	the plane shape, who throws it, how hard, where

Choose the right tool

Measure	Tool	Unit
length	ruler, tape measure	cm, m
temperature	thermometer	°C
time	stopwatch	seconds
force	forcemeter	newtons (N)



Read scales carefully.

Repeat it!

Do each test **three times**. If one result is very different, check it again. Repeating makes results more **reliable** (you can trust them more).

Present and explain

Put results in a **table**, then draw a **bar chart**. Look for a **pattern**. Then say what the evidence shows: was your prediction right?

Remember

Tables need **headings** and **units**, for example "Distance (m)".

Key words

evidence facts and measurements that show what is true

predict say what you think will happen and why

fair test change one thing, keep everything else the same

reliable results you can trust because you repeated them

pattern something that happens again and again in the results

Tools, units and tests

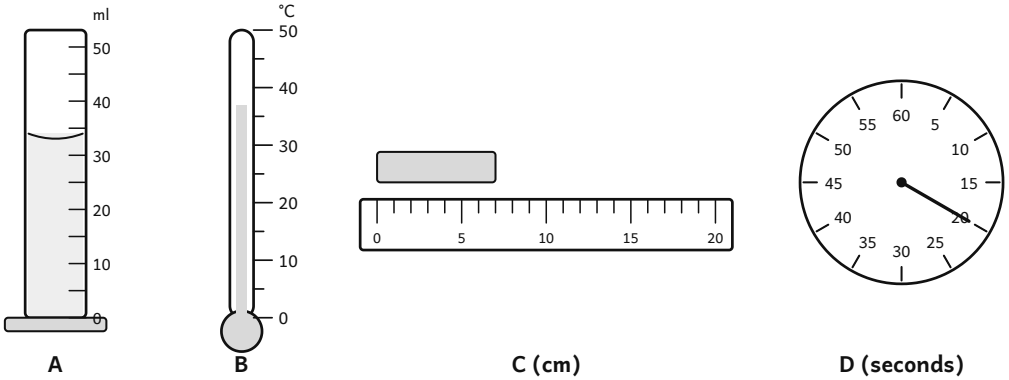
★☆☆
Grade 3

Name: Class: Date:

Learning goals

- Match tools and units.
- Spot testable questions.

1 Read each instrument. Include the unit. [4]



A B C (cm) D (seconds)

A	B	C	D

2 Draw a line to match the tool to the unit. [4]

Draw a line to match each pair, or write the letter.

thermometer ●	● **A** °C
stopwatch ●	● **B** cm
ruler ●	● **C** seconds
forcemeter ●	● **D** newtons

3 Can it be tested? Sort the questions. [4]

SORT THESE

Which ball bounces highest? **Does salt melt ice faster?**
Is red better than blue? **Which song is the nicest?**

Can be tested	Cannot be tested

THINK IT THROUGH · W1 · WORKING SCIENTIFICALLY

Making it fair

★★☆
Grade 3

Name: Class: Date:

Learning goals

- Plan a fair test.
- Explain repeats.

1 **Lina wants to know which cup keeps cha hot longest. She puts hot cha in a clay cup and cold cha in a steel cup. Is it fair? How can she fix it?** [3]

2 **Why should you repeat a measurement three times?** [2]

3 **Write a prediction with a reason: "Will a plant grow taller in sunlight or in shade?"** [2]

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Which paper makes the best paper plane?

★★☆ Grade 3

Name: Class: Date:

Learning goals

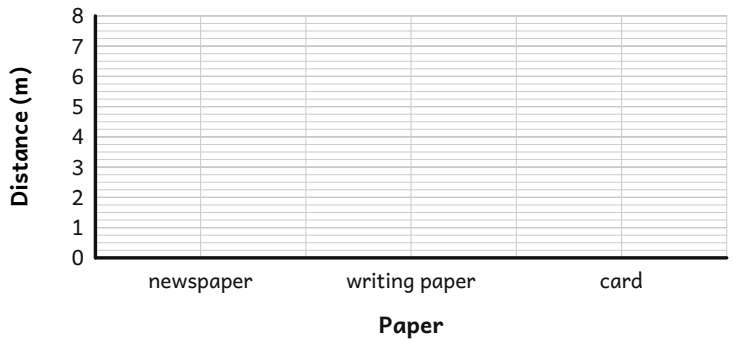
- Repeat measurements.
- Draw a bar chart.
- Say what the evidence shows.

Class 3A made the same paper plane from three kinds of paper and threw each one three times.

1 **Complete the table. Find the middle (median) result of each.** [3]

Paper	Try 1 (m)	Try 2 (m)	Try 3 (m)	Middle result (m)
newspaper	4	5	6	
writing paper	6	8	7	
card	3	2	4	

2 **Draw a bar chart of the middle results.** [3]



3 **Which paper was best? What does the evidence show?** [2]

.....

.....

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Testing umbrellas for the monsoon

★★★ Grade 3

Name: **Class:** **Date:**

Learning goals

- Plan a real test.
- Use evidence to choose.

Science · Design · Maths

The monsoon rains are coming. A shopkeeper in Rajshahi wants to sell the best umbrellas. He asks a Grade 3 class to test three kinds of cloth: cotton, plastic sheet and nylon.

The class drips 50 drops of water onto each cloth stretched over a cup, then measures how much water gets through.

A story about a class investigation. Results are example data.

1 **Complete the plan.** [3]

Change	
Measure	
Keep the same	

2 **Results: cotton 8 ml, plastic 0 ml, nylon 1 ml. Which cloth should the shopkeeper use? Why?** [2]

.....

.....

3 **What else should they test before choosing?** [1]

.....

.....

Show what you know

★★★ Grade 3

Name: Class: Date:

Learning goals

- Use science words.

1 Find the words.

K	E	K	R	E	P	E	A	T
F	T	E	B	Y	J	A	I	C
A	H	K	I	W	F	B	T	H
I	G	P	T	E	S	T	A	A
R	D	P	B	R	O	R	B	R
P	R	E	D	I	C	T	L	T
F	J	P	G	G	D	Y	E	T
H	M	E	A	S	U	R	E	J
U	T	H	M	W	A	L	W	S

chart fair measure predict repeat table test

2 Which tool measures time? [1]

◯ thermometer ◯ stopwatch

◯ ruler

3 In a fair test you change... [1]

◯ one thing ◯ two things

◯ everything

4 Challenge. Plan a test: "Which ball rolls furthest?" Write what you change, measure and keep the same. [3]

Change	
Measure	
Keep the same	

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
ask a question I can test	○	○	○
make a prediction with a reason	○	○	○
plan a fair test	○	○	○
measure with the right tool	○	○	○
repeat measurements	○	○	○
draw a bar chart	○	○	○